Project Proposal

DebtWise AI

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Introduction

Background Of Case Study

AI For financial literacy

This case study presents DebtWise AI, an AI-powered platform designed to help users manage debt and improve financial understanding, particularly in relation to **consumer credit** and **Buy Now, Pay Later (BNPL) services.** The project is connected to the theme of **AI for financial literacy** through the broader rise of digital finance. As digital payments become more common, short-term digital credit services such as BNPL are also growing, making financial decisions more frequent and more complex for users (OECD, 2025; OECD, 2026).

In this context, AI becomes relevant not only as a technical tool, but also as a means of helping users better understand debt, repayment behaviour, and financial consequences. DebtWise AI is therefore positioned as a practical response to the increasing use of digital financial services, where easier access to credit also creates a greater need for accessible and personalised financial guidance (OECD, 2026).

Chosen SDG

The primary Sustainable Development Goal addressed by this project is SDG 4: **Quality Education**, with a secondary connection to SDG 10: **Reduced Inequalities**. DebtWise AI functions as an educational tool that helps users understand debt, repayment strategies, and financial risk in a practical and personalised way. Simultaneously, it addresses the digital financial literacy gap by supporting users who may have access to digital financial services but lack the knowledge and confidence to use them effectively. In this sense, the project promotes lifelong learning while also helping reduce inequalities in digital financial capability.

Problem Statement

The project addresses a fundamental but important gap: While BNPL is expanding exponentially in Southeast Asia, users lack proper financial literacy. The gap is demonstrated by a higher adoption rate of BNPL services that are more prone to risk, especially among Gen Z users and low-income groups who are more likely to engage in impulse buying behavior (Khoirunisa et al., 2025; Milieu Insight, 2024; TechNode Global, 2024; Visa, 2023; Yulianita & Kurnia, 2025).

It is a phenomenon that is common throughout Southeast Asia. According to data from Global FinLit, Southeast Asia is one of the least financially literate regions globally. Fewer than 30% of adults are financially literate in some parts of the region. In Cambodia, financial literacy is at a dismal 18%, indicating a lack of understanding of basic concepts such as compound interest, inflation, and diversification of investments (Cole, 2022). Furthermore, a 3,000-strong survey conducted across six countries in Southeast Asia found that 43% of respondents save only 10% of their income while 54% are not actively investing (Milieu Insight, 2024).

According to a report by TechNode Global using UnaFinancial's data, by 2023, 20.6% of adults in Southeast Asia had already adopted BNPL services, with Singapore, Philippines, and Malaysia reaching 31.3%, 27%, and 24.4% adoption rates by 2023, respectively. The report projects that by 2027, BNPL spending in Southeast Asia is expected to reach $53.2 billion, with Indonesia alone being expected to contribute $16.8 billion to that amount—a 209% increase from 2024 projections. In fact, 76% of Asia-Pacific region online shoppers have already tried installment payments, while Gen Z and millennials are at a very high adoption rate of 76%, according to Visa's Green Shoots Radar.

The concern isn't the growth itself, but the growth in an environment where users aren't financially literate enough to use BNPL correctly. A number of research studies conducted in Indonesia and Malaysia have confirmed that financial literacy is linked to more prudent BNPL usage. Conversely, low financial literacy is linked to impulsive or compulsive spending and poor financial well-being (Khoirunisa et al., 2025; Osman et al., 2024; Yulianita & Kurnia, 2025). A study of Gen Z users in Indonesia confirmed that financial literacy reduces BNPL usage. However, the positive effect of financial literacy on BNPL usage might be mitigated if the user has a consumption-oriented lifestyle (Khoirunisa et al., 2025). A study conducted in Probolinggo confirmed that low financial literacy and access to BNPL contribute to over-indebtedness, poor spending control, and cash flow difficulties (Yulianita & Kurnia, 2025).

A qualitative study conducted in Malaysia confirmed that young users of BNPL spend less time reading the terms and conditions of BNPL agreements, budget less, and tend to engage in impulsive spending. This contributes to financial stress compared to older users (Osman et al., 2024). The basic concern is that users of BNPL in Southeast Asia have more digital access to installment credit than they have financial literacy to comprehend the implications of the fees and interest charged on BNPL agreements, the responsibilities of users to repay BNPL debts, and the implications of BNPL on their financial well-being. There is an obvious need for an innovation that will assist users of BNPL to better manage their debt and repayments. That's where the innovation of DebtWise AI comes in.

Objectives

1. Enhance users' financial literacy by enabling them to comprehend their debts' balances, interest rates, due dates, as well as the implications of their debt repayment strategies.

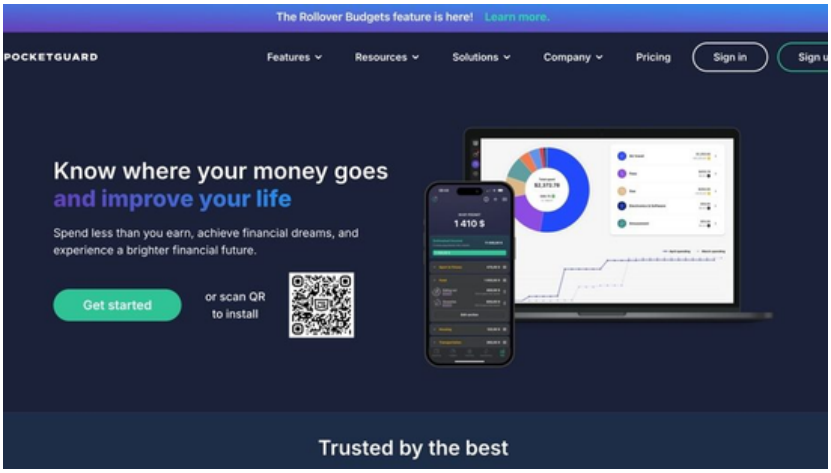
2. Facilitate responsible debt management by creating a single platform that monitors and organizes users' debts.

3. Facilitate more effective decision-making through recommendations generated by AI technology, simulations, as well as payoff strategy comparisons.

4. Minimize the risks associated with BNPL as well as digital short-term credit by making users more aware of their obligations.

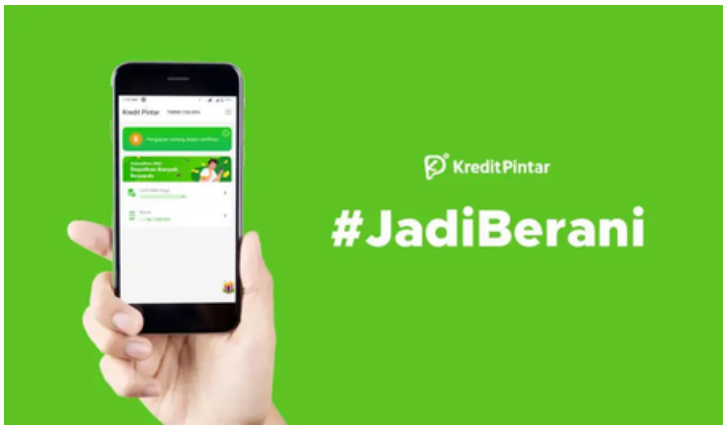
Review of Existing Solution

However, there are a few existing solutions that deal with a part of the problem, though none of them seem to be addressing all of it. Budgeting and debt tracking apps like PocketGuard, and NerdWallet are some of them. These apps assist users in managing their budget, keeping track of their bills, and monitoring their debts.



The second set of existing solutions includes AI-powered personal finance assistants like Cleo, which assist users in managing their finances by providing them with a chat interface for tracking their spending and setting goals. These are very useful in providing users with a better experience while accessing personal finance information and assistance.

In Southeast Asia, many popular fintech apps seem to be focusing on providing users with credit access rather than helping them understand their debts better. For example, Kredit Pintar is an online lending service that provides users with quick loans and offers information regarding their loan terms and conditions. It claims to be an AI-powered digital lending platform that offers users educational content through blogs. However, it is still a lending service and does not function as a debt management and personal finance education tool. Lending services also terrorize users if failed debt payment with extreme methods such as spreading to personal contacts.



The discussion above highlights a gap in existing personal finance and debt management apps and services, which are generally either focused on tracking finances, providing users with generic information, or offering users credit access. There are very few, if any, that provide users with all the necessary tools and information for addressing the case study problem. DebtWise AI is one of them, as it is designed for providing users with a better experience by utilizing the data it tracks for providing users with personal finance information and education, rather than just tracking their debts.

Methodology

Technical Track

DebtWise AI was developed as a full-stack web application that combines an interactive frontend, a REST-based backend, a relational database, and AI integration for personalised financial support. The system is designed to help users manage debt, simulate repayment strategies, and receive AI-assisted financial guidance in a structured and scalable way.

Tech Stack

The frontend was built using Next.js with React and TypeScript, providing a modern web interface with component-based development and strong type safety. Tailwind CSS and shadcn/ui were used to create a responsive and clean user interface, while Recharts was used for visualising debt distributions and repayment progress.

The backend was implemented using FastAPI, which was chosen for its lightweight structure, speed, and strong support for API development in Python. Supporting libraries included Pydantic for request and response validation, psycopg2 for PostgreSQL database access, python-jose for JWT-based authentication, and passlib with bcrypt for password hashing.

The database layer uses PostgreSQL, hosted through Supabase, to store user accounts, debt records, payment history, chatbot conversations, and monthly financial insights. This relational structure supports secure multi-user data handling and allows all financial records to be linked through foreign-key relationships.

For deployment, the project was designed to use Vercel for the frontend, Railway for the backend, and Supabase for the database. The AI chatbot component was integrated using Google Gemini 2.5 Flash through the Google Generative AI SDK.

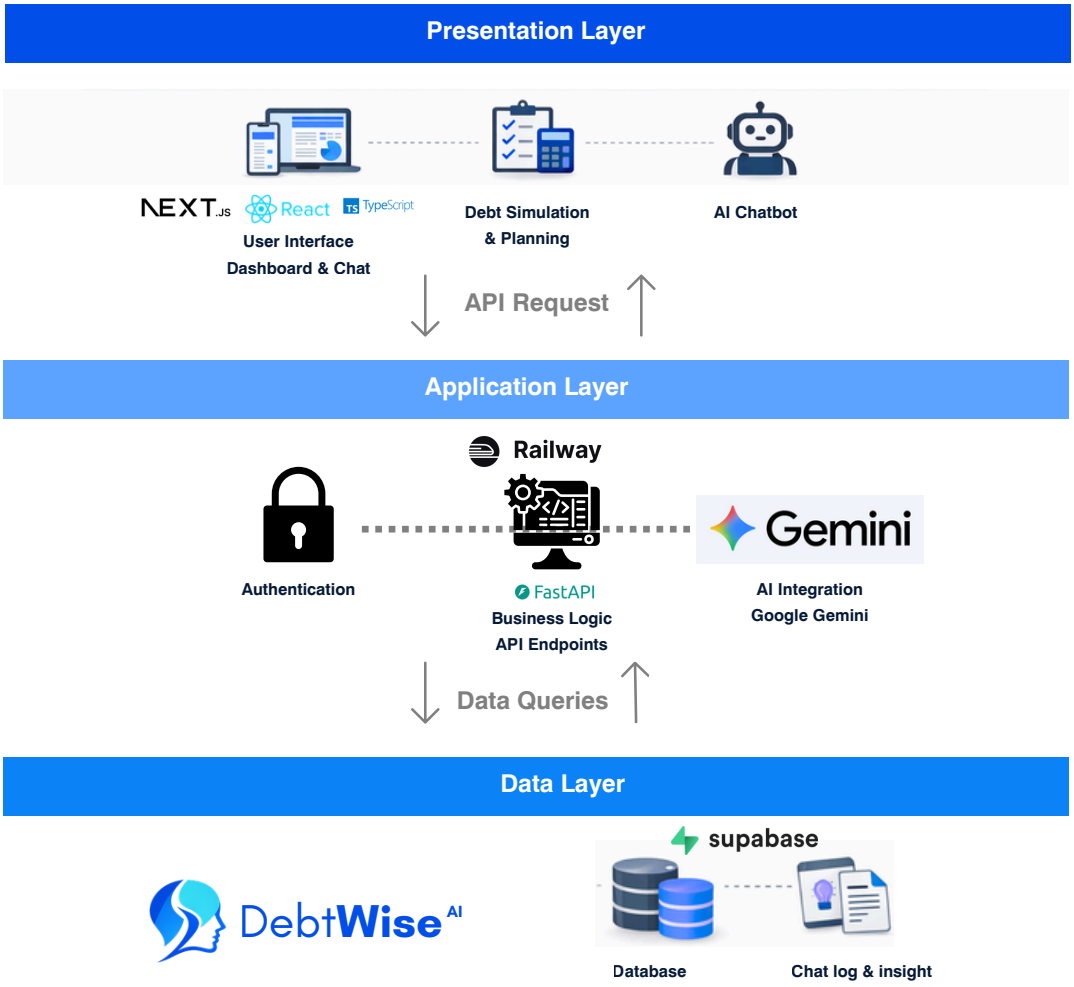
System Architecture

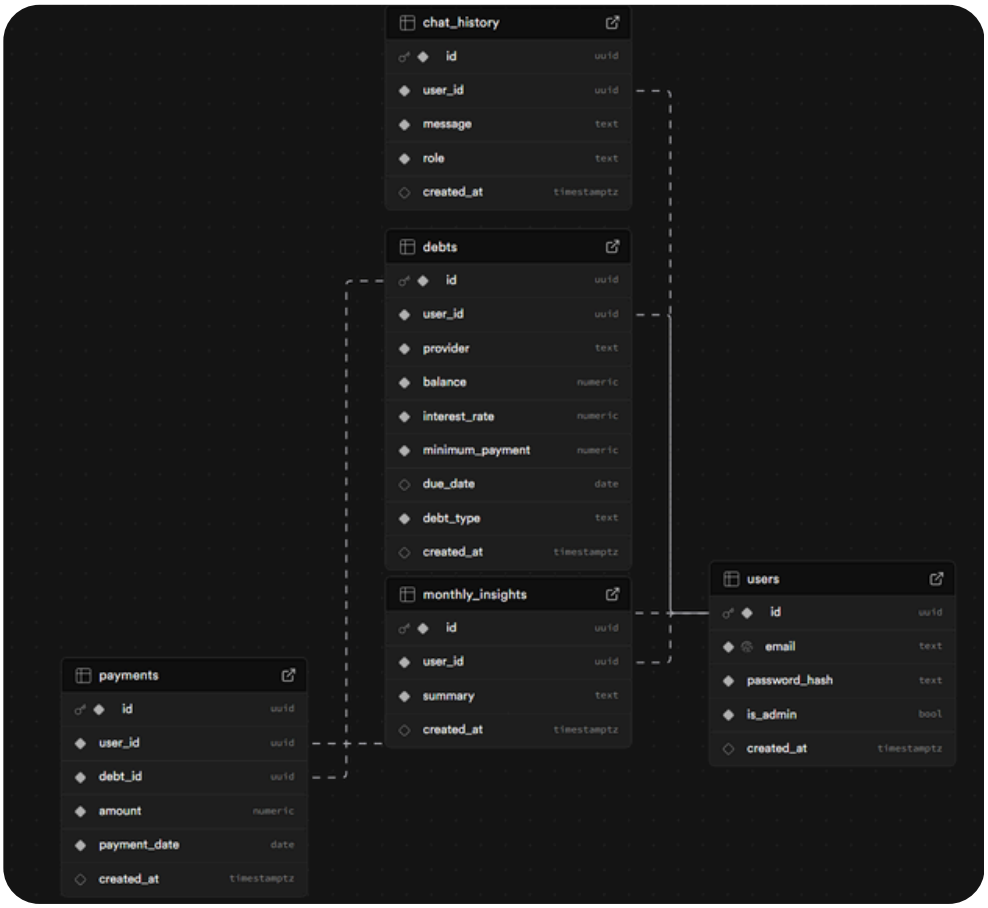
The architecture follows a three-tier structure consisting of the frontend, backend, and database.

At the presentation layer, the Next.js frontend handles user interaction, page rendering, authentication state, and dashboard visualisation. Users can log in, add debt entries, run simulations, generate repayment plans, and interact with the AI chatbot through this layer.

At the application layer, the FastAPI backend processes all business logic. It handles authentication, validates incoming requests, executes debt calculations, generates repayment plans, retrieves historical data, and connects to the AI service. The backend also exposes REST API endpoints such as /auth/login, /debts, /simulate, /plan, /chat, and /insights.

At the data layer, PostgreSQL stores structured data across several tables, including users, debts, payments, chat_history, and monthly_insights. Relationships between these tables ensure that each user's debt records, payment history, and AI-generated insights remain properly connected and isolated.





Database Scheme

The DebtWise AI data model is based on a relational PostgreSQL database hosted on Supabase. This data model organizes user financial data into linked tables, allowing the DebtWise AI system to efficiently store debt records, payments, chat history, and AI-generated insights in a structured format. Each user's data is kept separate and secure, while related data is connected through relational links between tables.

Users Table

The users table is the core table for user authentication and account data.

Key fields:

- id (uuid): This is a unique user ID.
- email (text): This is the user's login email.
- password_hash (text): This is the user's password hashed using bcrypt.
- is_admin (bool): This is a flag indicating whether the user has admin privileges.
- created_at (timestamptz): This is the date when the user account was created.

The users table acts as the main reference point for all other user-related tables.

Debts Table

The debts table contains all the debt information that a user wants to track.

Key fields:

- id (uuid): This is a unique ID for each debt.
- user_id (uuid): This is a reference back to the users table.
- provider (text): This is the lender or provider's name (e.g., Klarna or Afterpay).
- balance (numeric): This is the remaining balance on the debt.
- interest_rate (numeric): This is the interest rate applied.
- minimum_payment (numeric): This is the minimum amount that must be paid each month.
- due_date (date): This is the due date for the debt.
- debt_type (text): This is the type of debt (e.g., BNPL, Credit Card, Personal Loan, or Digital Loan).
- created_at (timestamptz): This is the date when the debt was added into the system.

The debts table represents the core financial dataset used by the user dashboard, simulator, repayment planner, and AI advisor.

Payments Table

The payments table contains all the payments made by the user toward their debts.

Key fields:

- id (uuid)
- user_id (uuid): The link to the user
- debt_id (uuid): The link to the specific debt
- amount (numeric): The amount being paid
- payment_date (date): The date the payment was made
- created_at (timestamptz): The date the payment record was created

This table allows the system to store repayment history and perform calculations related to repayment progress and financial analytics.

Chat History Table

The chat_history table keeps track of the conversation between the user and the AI advisor.

Key fields:

- id (uuid): The chat message identifier
- user_id (uuid): The link to the user
- message (text): The message sent by either the user or the AI advisor
- role (text): Identifies whether the message was sent by the user or the AI assistant
- created_at (timestamptz): The date the message was created

This table enables continuous AI conversations and allows users to review past discussions and recommendations from the AI advisor.

Monthly Insights Table

The monthly_insights table stores AI-generated financial reports summarizing user activity.

Key fields:

- id (uuid): The insight report identifier
- user_id (uuid): The link to the user
- summary (text): The AI-generated summary and analysis of financial activity
- created_at (timestamptz): The date the report was created

These insights help users review their financial activities and track progress over time.

Table Relationships

Foreign keys are used to maintain data consistency and establish relationships between the tables.

The relationships in the database are defined as follows:

- users → debts (one-to-many)
- users → payments (one-to-many)
- debts → payments (one-to-many)
- users → chat_history (one-to-many)
- users → monthly_insights (one-to-many)

These relationships ensure that the data remains organized, consistent, and properly connected across the system.

Implementation Process

The implementation process began with defining the core problem scope, namely helping users better understand and manage debt in a digital financial environment. Based on this, the system was divided into several core modules: debt dashboard, debt CRUD management, payment simulator, repayment plan generator, AI chatbot, monthly insights, and authentication.

The next stage involved designing the database schema. The schema was structured so that each user could own multiple debts, each debt could have multiple payments, and each user could also store chat history and monthly AI-generated reports. This relational structure made it possible to build the application features on a consistent data model.

After the schema design, the backend APIs were implemented first. Authentication endpoints were created for signup and login, followed by endpoints for debt management, repayment simulation, repayment strategy generation, AI chat, and insights generation. This step ensured that the application logic was functional before integrating it into the user interface.

The frontend implementation was then developed around these API endpoints. Pages were organised using the Next.js App Router, with separate views for login, signup, dashboard, debts, chat, simulator, plan, and insights. Reusable components and utility modules were added to support cleaner code structure and consistent interaction with the backend.
Finally, the AI integration was connected to the debt data so that chatbot responses and insights could become context-aware. Instead of giving generic responses, the AI component was designed to use the user's actual debt records to provide more personalised explanations and repayment suggestions.

Key Technical Features

Some of the main technical features implemented in the system include:

- JWT-based authentication with protected routes
- Debt CRUD operations connected to a PostgreSQL database
- Repayment simulation logic for comparing minimum payments with accelerated repayment
- Repayment plan generation using Snowball, Avalanche, and Cashflow strategies
- Streaming AI chatbot responses for real-time interaction
- Persistent chat history and monthly insights storage
- Responsive dashboard visualisation using charts and summary cards

BMC (Business Model Canvas)

The platform is named DebtWise AI. The platform is an artificial intelligence-based platform meant to assist individuals in managing their debts while improving their financial literacy, especially for users of BNPL services, Gen Z, and digitally savvy consumers in Southeast Asia who are prone to financial challenges due to their poor understanding of their credit obligations. The platform's value proposition is to offer debt management services via artificial intelligence. The platform's activities include web application development, maintenance of the web application, the provision of financial advice via artificial intelligence, provision of repayment strategies such as the snowball and avalanche methods, and financial education provision. The platform's key resources include the technology stack, artificial intelligence technology based on Google Gemini, the web application developers, database infrastructure based on Supabase, and financial data used to provide financial insights.

DebtWise AI partners with key players like Google for AI services, Vercel for frontend deployment, Railway for backend hosting, Supabase for database management, BNPL providers, financial institutions, and financial education organizations. The platform creates customer relationships with AI chatbots for support services, a self-service dashboard for debt management, smart simulations for repayment strategies, and educational engagement for understanding customer behavior. Customers use the platform primarily through the responsive web application with direct online access, AI integrations, digital marketing strategies, and partnerships with financial or educational institutions. The business model primarily targets BNPL users in Southeast Asia, Gen Z and millennials, people with low financial literacy, low-income earners, first-time credit users, people with multiple debts, or people with financial stress. The cost structure for the platform includes the salaries of the development team, infrastructure costs, database management costs, AI API costs, and storage costs. The revenue model for the platform is based on a freemium model with basic debt tracking available for free. Advanced AI advisory services, financial planning tools, and simulations will be available for a premium. The platform helps to fill the gap between easy access to digital credit and low financial literacy with the integration of financial education, AI technology, and practical debt management tools.

Results

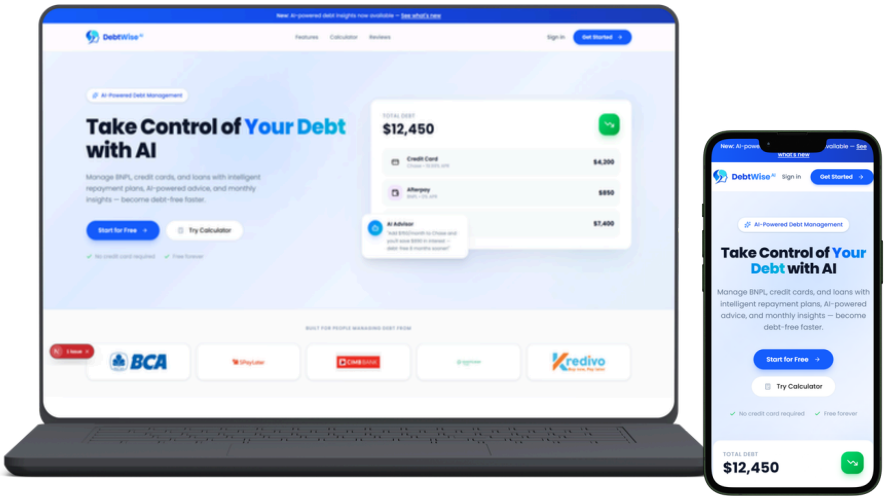
The project came together as a fully functional full-stack web application that incorporates debt management and AI financial advice. The application allows users to sign up and manage their debts. They can also use the application to simulate and plan their debt repayment strategies.

Additionally, the application provides users with AI advice on how to manage their debts. Therefore, the project was successful as it was meant to be. A digital platform that helps users manage their debts better was realized. The project also demonstrates that it is possible to develop a full-stack web application that provides users with an effective way of managing their debts. The application also demonstrates that it is possible to develop an AI chatbot that provides users with financial advice based on their debt situations. During the project development, the main features of the application functioned as expected. The features allow users to perform CRUD operations on their debts. They also allow users to perform simulations that compare the minimum payments of their debts and the payments they make to pay off their debts quickly.

Additionally, the features allow users to plan their debt repayment strategies using the snowball method, avalanche method, and cash flow method. The application also demonstrates that the main parts of the system function well together. The front-end and the FastAPI-based back-end communicate well. The FastAPI-based back-end also functioned well as it was able to handle the authentication of users and the AI chatbot.

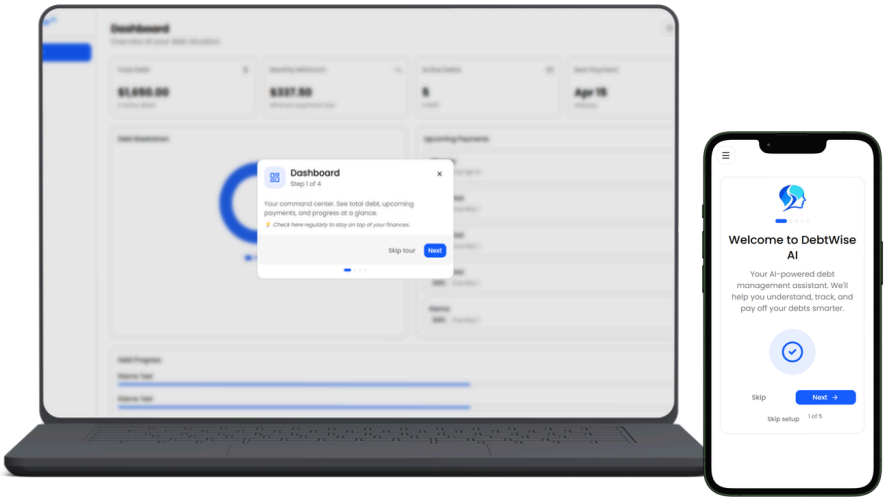
Finally, the PostgreSQL database functioned well as it was able to keep track of the user accounts, debts, payments, and AI chatbot advice. Therefore, the project was successful as it was meant to be. A digital platform that helps users manage their debts better was realized. The project also demonstrates that it is possible to develop a full-stack web application that provides users with an effective way of managing their debts. Additionally, the project demonstrates that it is possible to develop an AI chatbot that provides users with financial advice based on their debt situations.

Prototype Showcase



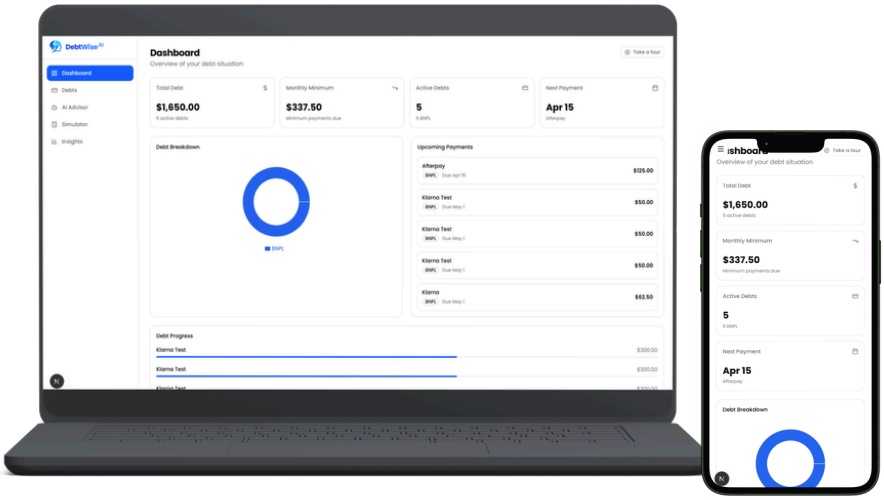
Landing Page

The home page is the first point of contact for users who are seeking information on the DebtWise AI platform. The purpose of this page is to inform users of the value that the platform offers, highlight its major features, and encourage users to take the first step in their debt management process.



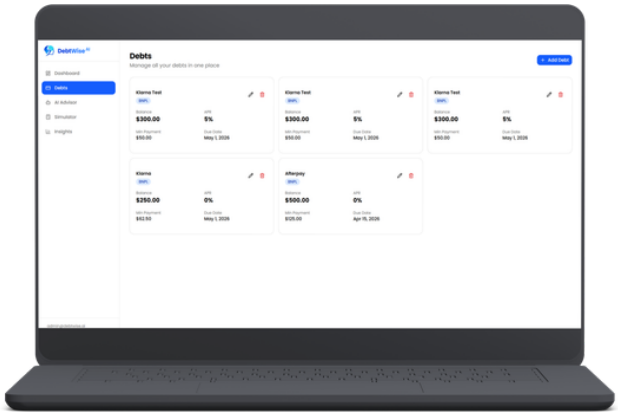
User Onboarding

The User Onboarding screen is the first interactive process that a new user will encounter when launching the DebtWise AI app. The primary goal of this process is to get the user acquainted with the platform and its importance while setting them up for effective management of their debts.

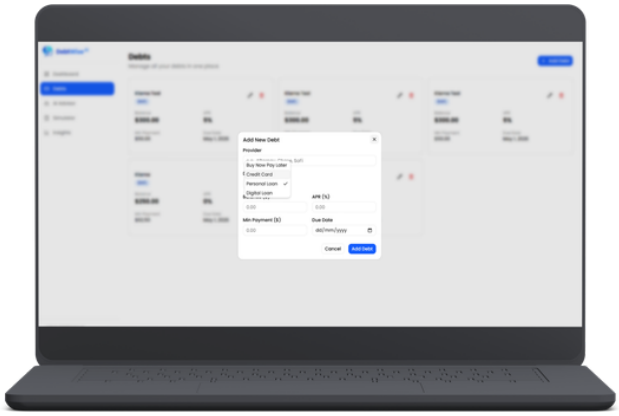


Main Dashboard

The Main Dashboard is the central point of the DebtWise AI system, allowing users to have an overview of their financial situation and one platform to manage all debts. The Main Dashboard is designed to provide users with valuable insights by allowing them to view condensed financial data, visualize the distribution of debts, and view upcoming payment obligations.



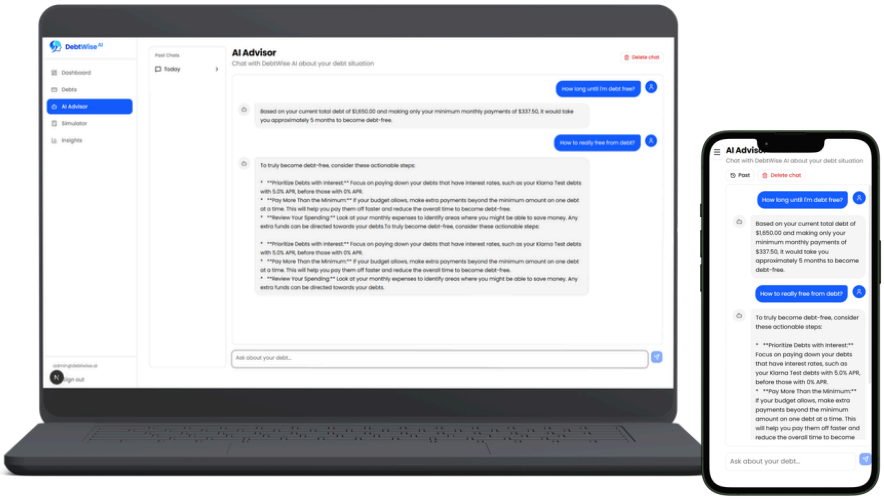
Main Debt Management Screen



Add Debt Screen

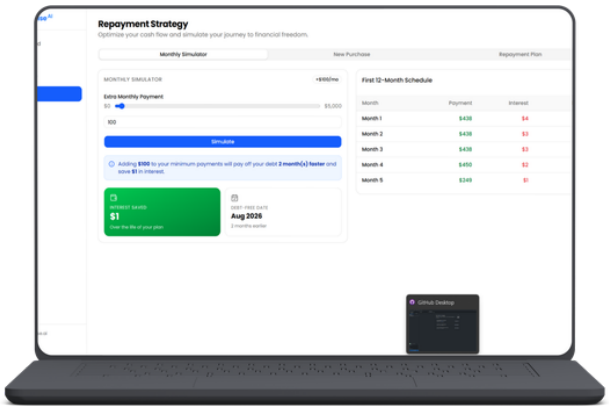
Debt Management

The Debt Management screen is where you can view your debts, add your debts, edit your debts, or delete your debts in a clean and organized format. It is essentially the main hub for all your debt records that the system maintains for you.

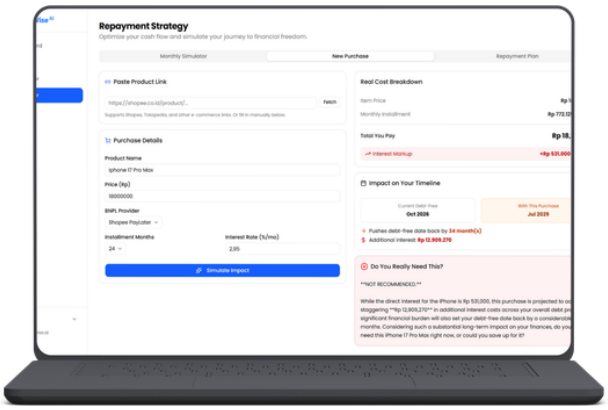


AI Chat Advisor

The feature of AI Advisor screen provides an interactive chat where you can talk directly with the DebtWise AI assistant for personalized advice on your debt situation. It uses AI to analyze your saved financial information and give you contextual recommendations with explanations.



Monthly Simulator



New Purchase Advisor

Debt Simulator Payment

The Debt Simulator and Repayment Plan module allows users to analyze how different financial decisions affect their debt repayment timeline. This feature helps users understand the long-term impact of their payments or purchases before making financial commitments.

Monthly Payment Simulator

The Monthly Simulator allows you to "explore what happens if you increase your monthly payments to accelerate your debt reduction." This can be done by sliding a control to increase the amount you pay on top of your regular amount.

When you do the simulation, the tool will display the following to you:
- The amount of interest you can save
- The number of months you can save
- The date you can achieve debt freedom

Additionally, the entire 12-month plan is shown to you, including the amount you are paying each month, the amount of interest you are paying each month, and the amount you have left to pay.

New Purchase Advisor

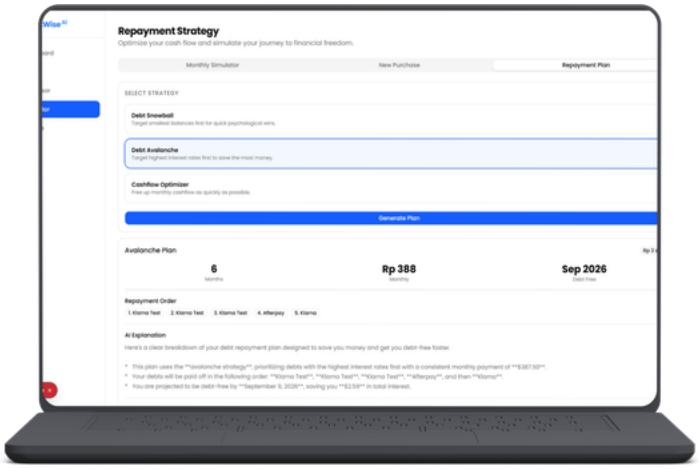
The New Purchase simulator helps you determine whether making another BNPL purchase will ruin your financial plans.

You can input the following:
- Product name
- Amount of the purchase
- BNPL provider
- Number of months for installment
- Monthly interest rate

The simulator then offers you a sales pitch with the following cost breakdown:
- Amount paid out
- Monthly installment
- Interest markup from the BNPL service provider

It also illustrates the impact on your financial freedom date. For example, it can demonstrate whether making another purchase extends your financial freedom date by a few months or increases your interest payments.

To aid your decision-making process, a warning panel appears to determine whether the purchase is financially wise based on the projected outcome.



Repayment Plan

Repayment Plan

DebtWise AI can create a clear plan of action, utilizing methods such as the Avalanche Method, where you focus on paying off your highest-interest loans first. It considers your overall debt situation and then comes up with a detailed plan that includes:
- A rough idea of when you can be debt-free
- A specific date when you can be debt-free
- The total interest you will be paying
- The interest you will save
- The order in which you should be paying your debts

There is also an AI-friendly explanation section that discusses why this is a good strategy and why this plan will work for you.

Repayment Plan (Exportable PDF)

For a more practical approach to using the tool, you can export the repayment plan as a PDF. The exported document will contain:
- A summary of the overall repayment strategy
- An estimated period to be debt-free
- Payment plan for the initial months
- Calculated interests and balances

The exported document can be used to show the lenders or debt collectors a plan to settle the debts. By providing a well-structured plan to settle the debts, the users can alleviate some of the stress from the debt collectors.

AI Acknowledgement

ChatGPT

For code debugging and Asset Generation

ChatGPT was used as an assistance tool during the development process to support research, documentation formatting, conceptual structuring, and technical explanations related to the project. These tools helped streamline the writing process and clarify certain implementation concepts.

Gemini

API Usage for the application

This project utilized artificial intelligence tools to support parts of the development and documentation process. Google Gemini was used as the primary AI model integrated within the application to provide financial explanations, repayment guidance, and contextual advice based on user debt data through the Google Generative AI SDK.

Conclusion

DebtWise AI shows how web tech and AI can come together in today's world to help people make smarter financial decisions and better manage their debt levels. It combines debt tracking, debt simulation, AI assistance, and financial information into a single platform, helping people understand their current financial state and move closer to a debt-free life.

It uses a full-stack approach with a Next.js frontend, a FastAPI backend, a PostgreSQL database, and Google Gemini AI, resulting in a robust platform for debt management. Users can track their debt, simulate different debt repayment options, analyze how different purchases will impact their financial state, receive personalized AI assistance, and generate a structured debt repayment plan.

It also provides useful features such as exporting a repayment plan in PDF format, allowing users to clearly communicate their debt repayment strategy with lenders and collectors, helping people manage their debt in a more effective manner.

Overall, the project was a success, as it developed an AI-powered debt management platform that enhances transparency, promotes responsible debt repayment, and improves financial literacy. It uses complex financial information and translates it into a clear and actionable format, helping people make better financial decisions and move closer to a debt-free life.

In terms of further development, it could be extended to connect with actual financial accounts, expand its analytics, and personalize its AI assistance based on actual financial behaviors.

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